

Field-Level Specifications

AI-Driven Prior Authorization System

Complete Field Definitions & Payer Integration Specifications

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Table of Contents

- Section 1: Provider Portal - Field Specifications
- Section 2: Clinical Reviewer Workbench - Field Specifications
- Section 3: Member Portal - Field Specifications
- Section 4: Administrative Dashboard - Field Specifications
- Section 5: Payer Integration Specifications
- Section 6: API Field Mappings
- Section 7: Data Validation Rules
- Section 8: Error Handling & Messages

Section 1: Provider Portal - Field Specifications

1.1 PA Request Submission Form

This form allows providers to submit new prior authorization requests.

1.1.1 Patient Information Section

Field Name	Field Type	Max Length	Required	Validation Rules	Example
First Name	Text	50	Yes	Alpha only, no special chars	Sarah
Last Name	Text	50	Yes	Alpha only, no special chars	Johnson
Middle Initial	Text	1	No	Alpha only, single character	M
Date of Birth	Date Picker	MM/DD/YYYY	Yes	Must be past date, age 0-120	03/15/1978
Gender	Dropdown	N/A	Yes	M, F, X, U (Unknown)	F
Member ID	Text	20	Yes	Alphanumeric, auto-validated	MB12345678
Insurance Group #	Text	15	No	Alphanumeric	GRP987654
Phone Number	Phone	14	Yes	US format (XXX) XXX-XXXX	(555) 123-4567
Email Address	Email	100	No	Valid email format	sarah.j@email.com
Address Line 1	Text	100	Yes	Street address	123 Main Street

1.1.2 Clinical Information Section

Field Name	Field Type	Max Length	Required	Validation Rules	Example
Primary Diagnosis	ICD-10 Lookup	10	Yes	Valid ICD-10 code with description	M54.5 - Low back pain
Secondary Diagnosis 1	ICD-10 Lookup	10	No	Valid ICD-10 code	M51.26 - Disc disorder
Secondary	ICD-10	10	No	Valid ICD-10 code	M79.1 -

Diagnosis 2	Lookup				Myalgia
Procedure/Service Code	CPT Lookup	10	Yes	Valid CPT/HCPCS code	72148 - MRI lumbar spine
Service Type	Dropdown	N/A	Yes	Predefined list	Diagnostic Imaging
Place of Service	Dropdown	N/A	Yes	Standard POS codes	Outpatient Hospital
Urgency Level	Radio Button	N/A	Yes	Routine/Urgent/Emergent	Routine
Estimated Cost	Currency	10	No	Numeric, 2 decimal places	\$1,250.00
Quantity/Units	Number	5	Yes	Integer, min 1	1
Frequency	Text	50	No	Free text	Once
Duration of Treatment	Text	50	No	Free text	1 day
Clinical Summary	Text Area	5000	Yes	Min 100 characters	Patient presents with...
Relevant Lab Results	Text Area	2000	No	Free text	WBC: 7.2, ESR: 15
Prior Treatments	Text Area	2000	Yes	Document conservative care	PT x 12 sessions, NSAIDs
Treating Physician Name	Text	100	Yes	Auto-populated from profile	Dr. Robert Smith

1.1.3 Document Upload Section

Field Name	Field Type	Max Length	Required	Validation Rules	Example
Clinical Notes	File Upload	25MB	Yes	PDF, JPG, PNG, TIFF	clinical_notes.pdf
Lab Results	File Upload	25MB	No	PDF, JPG, PNG, TIFF	labs_02092026.pdf
Imaging Reports	File Upload	25MB	No	PDF, JPG, PNG, TIFF, DICOM	xray_lumbar.pdf
Prescription	File Upload	25MB	No	PDF, JPG, PNG	prescription.pdf
Other Supporting Docs	File Upload	25MB	No	PDF, JPG, PNG, TIFF	additional_info.pdf

1.1.4 Provider/Facility Information Section

Field Name	Field Type	Max Length	Required	Validation	Example
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Rules					
Requesting Provider NPI	Text	10	Yes	Valid 10-digit NPI	1234567890
Provider Name	Text	100	Yes	Auto-populated	Dr. Robert Smith
Provider Specialty	Dropdown	N/A	Yes	Standard taxonomy codes	Orthopedic Surgery
Provider Tax ID	Text	20	Yes	Valid EIN format	12-3456789
Provider Phone	Phone	14	Yes	US format	(555) 987-6543
Provider Fax	Phone	14	Yes	US format	(555) 987-6544
Facility Name	Text	100	Yes	Auto-populated or manual	St. Mary's Medical Center
Facility NPI	Text	10	Yes	Valid 10-digit NPI	9876543210
Facility Address	Text	200	Yes	Full address	456 Hospital Blvd, City, ST

1.2 PA Status Tracking Dashboard

Displays all PA requests submitted by the provider with real-time status updates.

Field Name	Field Type	Data Source	Display Format	User Actions
PA Number	Read-only Text	System Generated	PA-YYYY-NNNNNN	Click to view details
Patient Name	Read-only Text	Patient record	Last, First M.	Click to view
Service Description	Read-only Text	CPT description	Full text	None
Status	Badge/Tag	Workflow engine	Color-coded label	None
Submission Date	Read-only Date	Audit log	MM/DD/YYYY HH:MM AM/PM	Sort/Filter
Decision Date	Read-only Date	Decision record	MM/DD/YYYY	Sort/Filter
Decision	Badge/Tag	Decision record	Approved/Denied/Pended	None
Assigned Reviewer	Read-only Text	Assignment system	Name or "AI Auto-Approved"	None
Days Pending	Calculated	System calculation	Integer + "days"	Sort
Urgency	Badge/Tag	Submission data	Routine/Urgent/Emergent	Filter
Auth Number	Read-only Text	Authorization system	Alphanumeric	Copy to clipboard
Valid Through	Read-only Date	Authorization record	MM/DD/YYYY	None
Download Letter	Button	Document storage	PDF icon + "Download"	Click to download
Appeal Status	Badge/Tag	Appeals system	No Appeal/In Appeal/Resolved	Click to view

Section 2: Clinical Reviewer Workbench - Field Specifications

2.1 Review Queue Dashboard

Field Name	Field Type	Max Length	Required	Validation Rules	Example
PA Number	Hyperlink	15	System	Click to open case	PA-2026-001234
Patient Name	Text	100	System	Display only, PHI masked per RBAC	Johnson, Sarah
Member ID (Masked)	Text	20	System	Last 4 digits only: ****5678	****5678
Service Type	Text	100	System	CPT description	MRI Lumbar Spine
Diagnosis	Text	150	System	Primary ICD-10 + description	M54.5 - Low back pain
Urgency	Badge	N/A	System	Color-coded: Red=Emergent, Orange=Urgent	Urgent
Deadline	Calculated	N/A	System	Hours or days remaining	6 hours
Received Date/Time	Date/Time	N/A	System	MM/DD/YYYY HH:MM AM/PM	02/09/2026 08:15 AM
Days in Queue	Calculated	N/A	System	Auto-calculated from received date	0.5 days
AI Confidence Score	Percentage	N/A	AI Engine	Color-coded: Green >90%, Yellow 70-90%, Red <70%	94%
AI Recommendation	Badge	N/A	AI Engine	Approve/Deny/Review Required	Approve
Assigned To	Text	N/A	System	Reviewer name or "Unassigned"	Dr. Emily Parker
Payer	Text	50	System	Insurance company name	UnitedHealthcare
Priority Score	Number	N/A	System	Calculated urgency + deadline weight	87
Documentation Complete	Checkbox	N/A	AI/System	Green check or red X	✓
Action Button	Button	N/A	User	Click to "Review Case"	Review

2.2 Case Detail Review Page

2.2.1 Patient Demographics Panel

Field Name	Field Type	Source	Editable	Display Format
Full Name	Text	Submission + Eligibility	No	Last, First Middle
Date of Birth	Date	Submission + Eligibility	No	MM/DD/YYYY (Age: XX)
Gender	Text	Submission + Eligibility	No	Male/Female/Other
Member ID	Text	Submission + Eligibility	No	Full ID visible to reviewer
Insurance Group	Text	Eligibility System	No	Group number + name
Payer Name	Text	Eligibility System	No	Full payer name
Coverage Status	Badge	Eligibility System	No	Active/Inactive/Pending
Effective Date	Date	Eligibility System	No	MM/DD/YYYY
Termination Date	Date	Eligibility System	No	MM/DD/YYYY or "N/A"
Primary Care Provider	Text	Payer System	No	PCP name + NPI
Contact Phone	Phone	Submission	No	(XXX) XXX-XXXX
Email	Email	Submission	No	member@email.com

2.2.2 Clinical Summary Panel

Field Name	Field Type	Source	Editable	Display Format
Primary Diagnosis	Text + Code	Submission + AI extraction	No	ICD-10: M54.5 - Low back pain
Secondary Diagnoses	List	Submission + AI extraction	No	Bullet list of ICD-10 codes
Requested Service	Text + Code	Submission	No	CPT: 72148 - MRI lumbar spine
Service Category	Badge	AI Classification	No	Diagnostic Imaging
Clinical Summary	Rich Text	Submission + AI extraction	No	Full narrative with highlighting
Chief Complaint	Text	AI extraction	No	Extracted from clinical notes
History of Present Illness	Text	AI extraction	No	AI-summarized timeline

Physical Exam Findings	Text	AI extraction	No	Relevant exam findings
Relevant Labs	Structured Data	AI extraction	No	Table of lab values
Relevant Imaging	Text	AI extraction	No	Prior imaging results
Medications	List	AI extraction	No	Current medications list
Allergies	List	AI extraction	No	Drug/food allergies
Prior Treatments	Timeline	AI extraction	No	Visual timeline of treatments
Treatment Duration	Text	AI extraction	No	Weeks/months of conservative care
Clinical Rationale	Text	Provider submission	No	Provider's justification

2.2.3 AI Analysis Panel

Field Name	Field Type	Source	Editable	Display Format
AI Recommendation	Badge	AI Engine	No	APPROVE/DENY/REVIEW REQUIRED
Confidence Score	Percentage + Gauge	AI Engine	No	94% with visual gauge
Risk Score	Number + Badge	AI Engine	No	Low/Medium/High risk
Rationale Summary	Rich Text	AI Engine	No	2-3 sentence explanation
Criteria Matched	Checklist	AI Engine	No	MCG/InterQual criteria met
Missing Criteria	Checklist	AI Engine	No	Red flags or gaps
Documentation Completeness	Percentage	AI Engine	No	95% complete
Missing Documentation	List	AI Engine	No	Items needed for full review
Supporting Evidence	Expandable List	AI Engine	No	Clinical guideline citations
Conflicting Factors	List	AI Engine	No	Factors arguing against approval
Similar Cases Analyzed	Number	AI Engine	No	2,847 cases
Similar Cases Outcome	Chart	AI Engine	No	Pie chart: 89% approved
Model Version	Text	AI Engine	No	v2.3.1
Processing Time	Duration	AI Engine	No	3.2 seconds

2.2.4 Clinical Guidelines Panel

Field Name	Field Type	Source	Editable	Display Format
Applicable Guideline	Text	AI + Rules Engine	No	MCG: Imaging for Low Back Pain
Guideline Version	Text	Guidelines DB	No	28th Edition, 2026
Criteria Text	Rich Text	Guidelines DB	No	Full criteria with highlighting
Must Meet (All)	Checklist	AI Analysis	No	Required criteria with status
Should Meet (Any)	Checklist	AI Analysis	No	Supporting criteria with status
Exclusion	Checklist	AI Analysis	No	Red flags

Criteria				checked
View Full Guideline	Button/Link	Guidelines DB	No	Link to full PDF/webpage

2.2.5 Decision/Action Panel

Field Name	Field Type	Required	Validation Rules	Notes
Decision	Radio Button	Yes	Must select one: Approve/Deny/Pend	Primary action
Clinical Notes	Text Area	No	Max 5000 chars, auto-save every 30 sec	Reviewer rationale
Denial Reason	Dropdown	If Deny	Must select from predefined list	Standardized reasons
Denial Sub-Reason	Dropdown	If Deny	Cascading based on reason	More specific code
Alternative Treatment	Text Area	If Deny	Max 1000 chars	Suggested alternatives
Pend Reason	Dropdown	If Pend	Must select reason	Why more info needed
Information Requested	Checklist	If Pend	Must select at least one	Specific items needed
Approved Quantity	Number	If Approve	Must be <= requested quantity	Units approved
Approved Duration	Number + Unit	If Approve	Days/weeks/months/visits	Auth validity period
Auth Start Date	Date Picker	If Approve	Cannot be past date	When auth begins
Auth End Date	Date Picker	If Approve	Must be after start date	When auth expires
Require MD Review	Checkbox	If Deny	Auto-checked for denials	Physician co-signature

Section 3: Member Portal - Field Specifications

3.1 Member Dashboard

Field Name	Field Type	Source	Editable	Display Format
PA Number	Hyperlink	PA System	No	PA-2026-001234
Service Description	Text	PA System	No	Plain language description
Provider Name	Text	PA System	No	Dr. Robert Smith
Status	Badge	PA System	No	Submitted/In Review/Approved/Denied
Submitted Date	Date	PA System	No	MM/DD/YYYY
Decision Date	Date	PA System	No	MM/DD/YYYY or "Pending"
Decision	Badge	PA System	No	Approved/Denied/More Info Needed
Authorization Number	Text	PA System	No	AUTH-123456 (if approved)
Valid Through	Date	PA System	No	MM/DD/YYYY (if approved)
View Letter	Button	Document System	No	Download PDF button

3.2 Appeal Submission Form

Field Name	Field Type	Max Length	Required	Validation Rules	Example
Original PA Number	Text (Auto)	15	Yes	Auto-populated from context	PA-2026-001234
Appeal Type	Radio Button	N/A	Yes	Standard (30 day) / Expedited (72 hr)	Expedited
Expedited Justification	Text Area	1000	If Expedited	Explain urgent medical need	Condition worsening...
Reason for Appeal	Text Area	5000	Yes	Min 100 characters	New clinical information...
Supporting Documents	File Upload	50MB total	No	PDF, JPG, PNG	new_mri_results.pdf
Provider Statement	File Upload	25MB	No	PDF signed by MD	provider_letter.pdf
Contact Phone	Phone	14	Yes	US format	(555) 123-4567
Preferred Contact Method	Dropdown	N/A	Yes	Phone/Email/Mail	Email

Section 4: Administrative Dashboard - Field Specifications

4.1 Analytics Dashboard Fields

Metric/Field Name	Calculation Method	Update Frequency	Display Type	Drill-Down Available
Total PAs Received	Count of all submissions	Real-time	Large number + trend	Yes - by date range
PAs in Queue	Count of pending status	Real-time	Number + urgency breakdown	Yes - view queue
Average TAT	Avg(decision_date - received_date)	Hourly	Hours + days	Yes - by service type
Auto-Approval Rate	Count(AI auto-approved) / Total * 100	Daily	Percentage + chart	Yes - by criteria
AI Accuracy	Count(AI=Reviewer) / Total reviewed * 100	Daily	Percentage + gauge	Yes - by case type
Appeal Rate	Count(appeals) / Total decisions * 100	Daily	Percentage + trend	Yes - view appeals
Overturn Rate	Count(overtured) / Total appeals * 100	Daily	Percentage	Yes - by reason
Approval Rate	Count(approved) / Total decisions * 100	Daily	Percentage + chart	Yes - by payer
Denial Rate	Count(denied) / Total decisions * 100	Daily	Percentage + chart	Yes - by reason
Pended Rate	Count(pended) / Total submissions * 100	Daily	Percentage	Yes - by reason
SLA Compliance	Count(within_deadline) / Total * 100	Real-time	Percentage + alert	Yes - violations
Reviewer Productivity	Cases per FTE per day	Daily	Number + leaderboard	Yes - by reviewer
Cost per PA	Total operating cost / # PAs	Monthly	Currency + trend	Yes - cost breakdown
Provider Satisfaction	NPS score from surveys	Monthly	Score (-100 to 100)	Yes - comments
System Uptime	Uptime minutes / Total minutes * 100	Real-time	Percentage + status	Yes - incidents

Section 5: Payer Integration Specifications

This section details integration requirements and field mappings for major payer systems. Each payer has unique data exchange requirements, API specifications, and business rules.

5.1 UnitedHealthcare (UnitedHealth Group)

5.1.1 Integration Overview

Integration Method	API (REST) + EDI X12 278
API Base URL	https://api.uhc.com/prior-auth/v2/
Authentication	OAuth 2.0 Client Credentials
Data Format	JSON (API) / X12 EDI (Batch)
Eligibility Verification	Separate API: /eligibility/v1/verify
Real-time Response	Yes (for urgent/expedited cases)
Batch Processing	Yes (nightly EDI 278 file exchange via SFTP)
SLA for Response	Urgent: 24hrs, Standard: 72hrs

5.1.2 UHC API Field Mappings

Our Field Name	UHC API Field	Data Type	Required	Special Notes
Member ID	subscriber.memberId	String(20)	Yes	UHC format: 3-letter prefix + 9 digits
First Name	subscriber.firstName	String(50)	Yes	Uppercase per UHC spec
Last Name	subscriber.lastName	String(50)	Yes	Uppercase per UHC spec
Date of Birth	subscriber.dateOfBirth	Date(YYY Y-MM-DD)	Yes	ISO 8601 format
Gender	subscriber.gender	String(1)	Yes	M/F/U only
Primary Diagnosis	clinicalInfo.primaryDiagnosis.code	String(10)	Yes	ICD-10 format without decimal
Diagnosis Description	clinicalInfo.primaryDiagnosis.description	String(200)	No	Auto-populated from code
Procedure Code	serviceRequest.procedureCode	String(10)	Yes	CPT or HCPCS Level II
Procedure	serviceRequest.modifiers	Array[String]	No	CPT modifiers if

re Modifier		ng]		applicable
Service Type	serviceRequest.serviceType	String(50)	Yes	UHC category codes
Place of Service	serviceRequest.placeOfService	String(2)	Yes	CMS POS codes (01-99)
Provider NPI	renderingProvider.npi	String(10)	Yes	Must be in UHC network
Provider Tax ID	renderingProvider.taxId	String(9)	Yes	EIN without dashes
Facility NPI	serviceLocation.npi	String(10)	Yes	Facility NPI if applicable
Request ed Start Date	serviceRequest.requestedStart Date	Date	Yes	Cannot be >30 days in past
Request ed Units	serviceRequest.requestedUnits	Integer	Yes	Min 1
Clinical Summar y	clinicalInfo.summary	String(5000)	Yes	Free text narrative
Urgency Level	authorizationRequest.priority	String	Yes	ROUTINE/URGENT/EMERGENCY
Attachm ent Referenc es	attachments	Array[Object]	No	Document IDs if uploaded separately
Submitt er Name	submitter.name	String(100)	Yes	Office/practice name
Submitt er Phone	submitter.phone	String(10)	Yes	Digits only, no formatting
Submitt er Fax	submitter.fax	String(10)	No	Digits only
Request ID	authorizationRequest.requestId	String(50)	Yes	Our internal PA number
Request Date	authorizationRequest.request Date	DateTime	Yes	ISO 8601 with timezone
Expe cted Respons e Date	N/A - UHC calculates	N/A	N/A	UHC returns in response

5.1.3 UHC Response Field Mappings

UHC Response Field	Our Field/Action	Data Type	Notes
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authorizationResponse.status	Decision Status	String	APPROVED/DENIED/PENDED/IN_REVIEW
authorizationResponse.authNumber	Authorization Number	String(20)	Store if APPROVED
authorizationResponse.effectiveDate	Auth Start Date	Date	When auth is valid from
authorizationResponse.expirationDate	Auth End Date	Date	When auth expires
authorizationResponse.approvedUnits	Approved Quantity	Integer	May differ from requested
authorizationResponse.denialReason	Denial Reason Code	String(10)	UHC standardized codes
authorizationResponse.denialDescription	Denial Reason Text	String(500)	Human-readable
authorizationResponse.pendReason	Pend Reason	String(500)	What info is needed
authorizationResponse.additionalInfo	Notes to Provider	String(2000)	Special instructions
authorizationResponse.reviewedBy	UHC Reviewer ID	String(50)	For audit trail

5.1.4 UHC Business Rules

- Member eligibility MUST be verified before submission (separate API call required)
- For OptumRx pharmacy benefits: use separate OptumRx API endpoint for medication PAs
- Imaging procedures require specific medical necessity documentation per UHC guidelines
- Out-of-network providers require additional justification and may auto-deny
- Behavioral health services route to Optum Behavioral Health - different API
- Transplant and specialty procedures require case manager assignment
- Appeals must be submitted within 60 days of denial date
- Peer-to-peer requests can be made via separate endpoint within 24hrs of denial

5.2 Aetna (CVS Health)

5.2.1 Integration Overview

Integration Method	FHIR R4 API + EDI X12 278
API Base URL	https://api.aetna.com/fhir/r4/
Authentication	OAuth 2.0 + SMART on FHIR
Data Format	FHIR Bundle resources
Eligibility Verification	FHIR Coverage resource
Real-time Response	Yes (CoverageEligibilityResponse)
Batch Processing	Yes (EDI 278 via SFTP)
SLA for Response	Urgent: 24hrs, Standard: 5 business days

5.2.2 Aetna FHIR Resource Mappings

Our Field Name	UHC API Field	Data Type	Required	Special Notes
Member ID	Patient.identifier[type=MB].value	String	Yes	Aetna member ID
First Name	Patient.name[0].given[0]	String	Yes	FHIR HumanName
Last Name	Patient.name[0].family	String	Yes	FHIR HumanName
Date of Birth	Patient.birthDate	Date	Yes	YYYY-MM-DD
Gender	Patient.gender	Code	Yes	male/female/other/unknown
Primary Diagnosis	Condition.code.coding[0].code	CodeableConcept	Yes	ICD-10 with system URL
Procedure Code	ServiceRequest.code.coding[0].code	CodeableConcept	Yes	CPT with system URL
Service Category	ServiceRequest.category	CodeableConcept	Yes	SNOMED CT preferred
Priority	ServiceRequest.priority	Code	Yes	routine/urgent/asap/stat
Provider NPI	Practitioner.identifier[type=NPI].value	String	Yes	10-digit NPI
Organization NPI	Organization.identifier[type=NPI].value	String	Yes	Facility NPI
Coverage Info	Coverage.id reference	Reference	Yes	Link to Coverage resource
Supporting Info	ServiceRequest.supportingInfo	Array[Reference]	No	References to DocumentReference
Clinical	DocumentReference.content.attachment	Base64	Yes	Embedded or

Notes	chment.data			referenced
Reason Code	ServiceRequest.reasonCode	CodeableConcept	Yes	Clinical justification
Requested Period Start	ServiceRequest.occurrence[x].start	DateTime	Yes	When service needed
Requested Period End	ServiceRequest.occurrence[x].end	DateTime	No	Duration if applicable
Quantity	ServiceRequest.quantityQuantity	Quantity	Yes	Units with UCUM code
Requester	ServiceRequest.requester	Reference	Yes	Reference to Practitioner
Intent	ServiceRequest.intent	Code	Yes	Always "order" for PA

5.2.3 Aetna Business Rules

- FHIR Bundle must include: Patient, Coverage, Condition, ServiceRequest, Practitioner resources
- CVS Caremark handles all pharmacy PAs - separate integration required
- Behavioral health routed to Aetna Behavioral Health - use specialty code BH
- Medical policy lookup available via separate API: /MedicalPolicy/v1/search
- Pre-service vs. post-service determination impacts routing and SLA
- Aetna requires attestation that patient has been informed of PA requirement
- Clinical attachments must be PDF, max 10MB per file, max 10 files
- Auto-approval available for certain preventive services and generic medications
- Network status checked automatically - out-of-network requires medical director review

5.3 CVS Health - Caremark (Pharmacy)

5.3.1 Integration Overview

Integration Method	NCPDP Telecom + Proprietary API
API Base URL	https://api.caremark.com/pharmacy-pa/v3/
Authentication	API Key + HMAC signature
Data Format	JSON with NCPDP field mappings
Formulary Check	Real-time formulary API: /formulary/v2/check
Real-time Response	Yes (for electronic PA - ePA)
Batch Processing	Yes (NCPDP batch files)
SLA for Response	Standard: 72hrs, Urgent/Expedited: 24hrs

5.3.2 CVS Caremark Field Mappings (Pharmacy PA)

Our Field Name	UHC API Field	Data Type	Required	Special Notes
Member ID	patient.memberId	String(20)	Yes	Caremark member ID (RxBin/RxPCN/RxGroup)
RxBin	patient.rxBin	String(6)	Yes	Bank Identification Number
RxPCN	patient.rxPcn	String(10)	Yes	Processor Control Number
RxGroup	patient.rxGroup	String(15)	Yes	Group identifier
Patient Name	patient.name	String(100)	Yes	First Last format
Date of Birth	patient.dateOfBirth	Date	Yes	YYYY-MM-DD
Gender	patient.gender	String(1)	Yes	M/F
Medication NDC	medication.ndcCode	String(11)	Yes	11-digit NDC (with dashes removed)
Medication Name	medication.productName	String(100)	Yes	Brand or generic name
Strength	medication.strength	String(50)	Yes	e.g., "20mg"
Dosage Form	medication.dosageForm	String(50)	Yes	Tablet, capsule, injection, etc.
Quantity	prescription.quantity	Decimal	Yes	Number of units
Days Supply	prescription.daysSupply	Integer	Yes	30, 60, 90 typical
Directions	prescription.directions	String(500)	Yes	SIG - prescriber instructions
Refills	prescription.refills	Integer	Yes	Number of refills authorized

Diagnosis Code	clinicalInfo.diagnosisCode	String(10)	Yes	ICD-10 for indication
Prescriber NPI	prescriber.npi	String(10)	Yes	Prescribing physician NPI
Prescriber Name	prescriber.name	String(100)	Yes	Dr. First Last
Prescriber Phone	prescriber.phone	String(10)	Yes	Digits only
Pharmacy NPI	pharmacy.npi	String(10)	Yes	Dispensing pharmacy NPI
Pharmacy NCPDP	pharmacy.ncdpdId	String(7)	Yes	Pharmacy chain code
Prior Medications Tried	clinicalInfo.priorMedications	Array[String]	Yes	Step therapy documentation
Clinical Rationale	clinicalInfo.justification	String(5000)	Yes	Why medication is medically necessary

5.3.3 CVS Caremark Business Rules

- Formulary check REQUIRED before PA submission - use /formulary/v2/check endpoint
- Step therapy: Must document failure/intolerance of preferred alternatives
- Quantity limits: Check MaxQuantity field in formulary response
- Specialty medications (>\$600/month): Route to Caremark Specialty division
- Prior authorization for brand when generic available: Requires generic trial documentation
- Biosimilar preference program: Must try biosimilar before brand biologic
- Age/gender edits: Some medications require specific age/gender criteria
- Duplicate therapy check: System flags if patient on similar medication
- Auto-approval for refills: If original PA approved and within validity period
- Clinical guidelines embedded in response: Use recommendedAlternatives field

5.4 Cigna Health

5.4.1 Integration Overview

Integration Method	Proprietary REST API + X12 278
API Base URL	https://api.cigna.com/authorization/v2/
Authentication	OAuth 2.0 + mTLS
Eligibility Check	/member/v1/eligibility
Real-time Decisions	Yes (for select procedures)
Batch Processing	Yes (EDI nightly)
SLA	Urgent: 24hrs, Standard: 48hrs

5.5 Humana

5.5.1 Integration Overview

Integration Method	HL7 v2.5 + X12 278 EDI
API Base URL	https://availability.humana.com/pa/
Authentication	Availity portal credentials + API key
Eligibility Check	Real-time via Availity
Real-time Decisions	Limited (Gold Card providers only)
Batch Processing	Yes (preferred method)
SLA	Standard: 5 business days

5.6 Blue Cross Blue Shield (BCBS)

NOTE: BCBS is a federation of 35+ independent companies. Integration varies by plan. This section covers common Anthem BCBS requirements (largest member).

Integration Method	Availity platform + Plan-specific APIs
API Base URL (Anthem)	https://api.anthem.com/prior-auth/v1/
Authentication	Availity SSO + OAuth 2.0
Eligibility Check	BlueCard system for cross-plan eligibility
Real-time Decisions	Yes (via eviCore for imaging/cardiology)
Batch Processing	Yes (X12 278)
Special Routing	eviCore, AIM, NIA for specialty services
SLA	Varies by plan: 24hrs-5 days

Section 6: Cross-Payer API Field Mappings

This section provides a consolidated view of how common PA fields map across different payer APIs, enabling efficient multi-payer integration with a single internal data model.

Internal Field	UHC API	Aetna FHIR	CVS Caremark	Cigna API	Common Data Type
member_id	subscriber.memberId	Patient.identifier.value	patient.memberId	member.id	String (20)
first_name	subscriber.firstName	Patient.name.given	patient.firstName	member.firstName	String (50)
last_name	subscriber.lastName	Patient.name.family	patient.lastName	member.lastName	String (50)
dob	subscriber.dateOfBirth	Patient.birthDate	patient.dateOfBirth	member.birthDate	Date(ISO)
gender	subscriber.gender	Patient.gender	patient.gender	member.sex	Code(M/F/U)
primary_dx	clinicalInfo.primaryDiagnosis.code	Condition.code	clinicalInfo.diagnosisCode	diagnosis[0].code	ICD-10
procedure_code	serviceRequest.procedureCode	ServiceRequest.code	medication.ndcCode	service.procedureCode	CPT/NDC
provider_npi	renderingProvider.npi	Practitioner.identifier	prescriber.npi	provider.npi	String (10)
urgency	authorizationRequest.priority	ServiceRequest.priority	request.urgencyLevel	request.urgency	Enum
clinical_notes	clinicalInfo.summary	DocumentReference	clinicalInfo.justification	clinical.narrative	Text(5000)
requested_date	serviceRequest.requestedStartDate	ServiceRequest.occurrence	prescription.fillDate	service.startDate	Date
quantity	serviceRequest.requestedUnits	ServiceRequest.quantity	prescription.quantity	service.quantity	Integer
place_of_service	serviceRequest.placeOfService	ServiceRequest.location	N/A (pharmacy)	service.location	Code(POS)
auth_number	authorizationResponse.authNumber	ClaimResponse.preAuthRef	response.approvalNumber	authorization.number	String (20)
decision	authorizationResponse.status	ClaimResponse.outcome	response.decision	decision.status	Enum
effective	authorizationRes	ClaimResponse.addl	response.eff	authorization	Date

e_date	ponse.effectiveDate	tem.servicedDate	ectiveDate	.startDate	
expiration_date	authorizationResponse.expirationDate	ClaimResponse.addItem.servicedPeriod.end	response.expirationDate	authorization.endDate	Date
denial_reason	authorizationResponse.denialReason	ClaimResponse.error	response.denialReason	decision.reason	Code
approved_units	authorizationResponse.approvedUnits	ClaimResponse.addItem.quantity	response.approvedQuantity	authorization.approvedUnits	Integer
reviewer_id	authorizationResponse.reviewedBy	ClaimResponse.reviewer	response.reviewerId	decision.reviewer	String (50)

Section 7: Data Validation Rules

7.1 Field-Level Validation Rules

Field Name	Validation Rule	Error Message	Error Code
Member ID	Regex: ^[A-Z0-9]{8,20}\$	Invalid Member ID format	VAL-001
Date of Birth	Must be past date, age 0-120	Invalid date of birth	VAL-002
Phone Number	US format: (XXX) XXX-XXXX	Invalid phone number	VAL-003
Email	RFC 5322 email validation	Invalid email address	VAL-004
ICD-10 Code	Valid ICD-10-CM code in current year	Invalid diagnosis code	VAL-005
CPT Code	Valid CPT/HCPCS in current year	Invalid procedure code	VAL-006
NPI	Valid 10-digit NPI, Luhn check	Invalid NPI	VAL-007
Tax ID	Valid EIN format XX-XXXXXXX	Invalid Tax ID	VAL-008
Zip Code	US format: XXXXX or XXXXX-XXXX	Invalid zip code	VAL-009
File Size	Max 25MB per file	File exceeds size limit	VAL-010
File Type	Allowed: PDF, JPG, PNG, TIFF	Unsupported file type	VAL-011
Quantity	Integer > 0, max 999	Invalid quantity	VAL-012
Clinical Summary	Min 100 characters, max 5000	Clinical summary too short/long	VAL-013
Date Range	Start date < End date	Invalid date range	VAL-014
Future Date Check	Cannot be >30 days in future	Date too far in future	VAL-015
Required Field	Cannot be null or empty	Required field missing	VAL-016
Cross-Field: Age/Service	Age appropriate for service	Service not age-appropriate	VAL-017
Duplicate Check	No duplicate PA for same service in 30 days	Duplicate PA exists	VAL-018
Network Check	Provider must be in-network	Out-of-network provider	VAL-019
Eligibility Check	Member must have active coverage	Member not eligible	VAL-020

Section 8: Error Handling & Messages

8.1 User-Facing Error Messages

Error Code	Scenario	User Message	Suggested Action
ERR-100	Member not found in eligibility system	We cannot verify this member ID. Please check the ID and try again.	Verify Member ID with insurance card
ERR-101	Member coverage inactive/terminated	This member's coverage is not currently active. Please contact the payer.	Check coverage effective dates
ERR-102	Duplicate PA submission	A prior authorization for this service already exists (PA-XXXX).	Review existing PA or wait for decision
ERR-103	Provider not in network	This provider is out-of-network. PA may require additional documentation.	Verify network status or provide justification
ERR-104	Missing required clinical documentation	Please upload clinical notes supporting medical necessity.	Upload required documents
ERR-105	File upload failed	Document upload failed. Please try again or use a smaller file.	Check file size (<25MB) and format
ERR-106	Invalid ICD-10 code	The diagnosis code entered is not valid for the current year.	Verify ICD-10 code or select from lookup
ERR-107	Service not covered	This service is not covered under the member's plan.	Review plan benefits or contact payer
ERR-108	Prior treatment required	Plan requires documented trial of [TREATMENT] before this service.	Document prior conservative care
ERR-109	Session timeout	Your session has expired. Please log in again.	Re-authenticate and resume
ERR-110	System unavailable	The PA system is temporarily unavailable. Please try again shortly.	Wait and retry, or submit via fax
ERR-111	Payer API error	Unable to reach	Request will process

		[PAYER] systems. Your request has been queued.	when connection restored
ERR-112	Age restriction violation	This service has age restrictions. Member must be [AGE RANGE].	Verify member age or select alternative service
ERR-113	Quantity exceeds plan limit	Requested quantity exceeds plan maximum of [X] units.	Reduce quantity or provide clinical justification
ERR-114	Missing peer-to-peer	Payer requires peer-to-peer discussion before denial appeal.	Contact payer to schedule peer-to-peer

8.2 API Integration Error Codes

Error Code	Scenario	User Message	Suggested Action
API-001	Authentication failure	Invalid OAuth token or credentials	Refresh token and retry
API-002	Rate limit exceeded	Too many requests to payer API	Implement exponential backoff
API-003	Malformed request	Request body does not match API schema	Validate request against schema
API-004	Payer system timeout	No response from payer within 30 seconds	Retry with exponential backoff
API-005	Unknown payer response	Payer returned unexpected response structure	Log for manual review
API-006	Certificate expired	mTLS certificate expired or invalid	Renew certificate
API-007	Unsupported payer version	Payer API version no longer supported	Upgrade integration to new version
API-008	Field mapping error	Required payer field missing in our data	Check field mapping configuration
API-009	Character encoding issue	Special characters not properly encoded	Apply UTF-8 encoding
API-010	Webhook delivery failed	Unable to deliver status update to provider	Queue for retry

— *End of Field-Level Specifications Document* —